Carbon 158 (2020) 827–835

Contents lists available at ScienceDirect

Carbon

journal homepage: www.elsevier.com/locate/carbon

C_{568}: A new two-dimensional sp^2-sp^3 hybridized allotrope of carbon

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ARTICLE INFO

Article history:
Received 4 September 2019
Received in revised form
13 November 2019
Accepted 21 November 2019
Available online 26 November 2019

ABSTRACT

Using first-principles calculations, we designed a new two-dimensional sp^2–sp^3 hybridized allotrope of carbon. This allotrope is made up of 5-6-8 carbon rings thus, known as C_{568}. C_{568} is an indirect semiconductor with a band gap of 1.13 eV (HSE06) at Γ-M symmetric point in the Brillouin zone. The possibility of the existence of C_{568} is evident from its energetic, dynamical, mechanical, and thermal stabilities. C_{568} maintains its structural integrity up to 800 K, which we confirmed by employing ab-initio molecular dynamics (AIMD) simulations. C_{568} shows extraordinary room temperature in-plane electronic mobility of order ~10^4 $cm^2V^{-1}s^{-1}$, which is significantly higher than the black phosphorus (~10^3 $cm^2V^{-1}s^{-1}$) monolayer. Energetically, this sheet is more stable than the other sp^2–sp^3-hybridized carbon allotropes such as pentagraphene and tetrahexcarbon. Moreover, C_{568} can be stacked into three different configurations, which also exhibit interesting electronic as well as transport properties. These configurations are also dynamically stable and show in-plane mobilities of order ~10^4 $cm^2V^{-1}s^{-1}$.

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1. Introduction

Carbon exists in many allotropic forms due to its ability to make *sp*, *sp*–sp^2, sp^2, sp^3, and sp^2–sp^3-hybridized bonding network. This ability of carbon lead to the formation of diamond, graphite, graphene [1–3], graphyne [4–6], graphdiyne [7,8], graphenylene [9], radialenes [10,11], pentaheptite [12], haeckelites [13], pentahexoctite [14], pentagraphene [15], phagraphene [16], tetrahexcarbon [17], T-graphene [18], and many other carbon allotropes. Among these, graphene is unique due to its remarkable properties such as high electronic mobility, good electrical and thermal conductivity, high modulus of elasticity and large surface area [2,19]. Moreover, graphene is ultimately thin with high mechanical strength making it attractive for a wide range of applications in smaller and faster electronics [20,21]. As a consequence, graphene has been the subject of interest in physics, chemistry, materials science, and biomedical related research fields [3].

However, the zero band gap in graphene limits its use in field-effect transistors (FETs) by restricting high on/off current ratio [22–24]. Similarly, the materials with relatively low charge carrier mobilities also limit their applications in high-performance electronic devices [25,26]. These challenges motivate further exploration of materials that at the same time shows a finite band gap, high charge carrier mobilities, and good stability. In this paper, we attempted to design a new carbon-based 2D material composed of 5, 6, and 8-membered rings of carbon, thus known as C_{568}. This is made up of three and four coordinated carbon atoms represented by C1 (sp^2) and C2 (sp^3), respectively. C_{568} is a semiconducting material with indirect-band gap of 1.13 eV (HSE06) at Γ-M symmetric points. It exhibits energetic, dynamical, mechanical and thermal stability. Energetically, C_{568} is more stable compared to other carbon allotropes such as T-graphene [18], T-carbon [27], pentagraphene [15], and recently proposed tetrahexcarbon [17]. Interestingly, C_{568} also shows high electronic mobility at room temperature (300 K) compared to black phosphorus monolayer [28].

In addition to C_{568}, structural, vibrational, electronic, and transport properties of its bilayers are also discussed. The C_{568} bilayers also show extraordinary electron mobilities at room temperature along with their energetic and dynamical stabilities. We also investigate the band gap tunability of C_{568}-bilayers under applied normal compressive strain due to its experimental feasibility and found that the indirect nature of band gap is insensitive towards the applied strain. The paper is organized as follows: the methodology applied in this study are stated in Section 2. Results and discussion of C_{568} and its bilayers based on their structures, stabilities (energetic, dynamic, mechanical, and thermal), electronic, and transport properties are given in Section 3. Finally, conclusions are given in Section 4.

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https://doi.org/10.1016/j.carbon.2019.11.062
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2. Method section

Our structure prediction technique is based on a combination of multi-start local search approach and atomistic simulation using Tersoff potential [29,30] followed by structure optimization using first-principles density functional theory (DFT) [31] from Vienna *ab initio* simulation package (VASP) [32,33]. We used projector augmented wave (PAW) [34,35] pseudopotentials method to take care of electron-ion interactions. For accurate band gap calculation, we used Heyd-Scuseria-Ernzerhof (HSE06) hybrid functional [32,36–38]. An adequate vacuum of 20 Å in *z*-direction and a well-converged 11 × 11 × 1 Monkhorst-Pack (MP) **k**-point mesh [39] for Brillouin zone sampling were used. For structural optimization, we used conjugate-gradient method until the absolute value of the components of the Hellman-Feynman forces were converged within 0.005 eV/Å. The convergence with respect to the energy cut-off of 400 eV was found to be sufficient. In C_{568} bilayers, the effect of van der Waals (vdW) interactions between layers were included using Grimme's DFT-D3 method [40]. For further confirmation of thermal stability, we applied ab-initio molecular dynamics (AIMD) simulations for 3 ps (time step of 1 fs) in 3 × 3 × 1 supercell at 300 K, and 800 K, respectively.

3. Results and discussion

C_{568} is predicted by the combination of multi-start local search approach and atomistic simulation techniques using Tersoff potential [29,30] with further structure optimization using first-principles calculations [31–33]. The unique network of 5, 6, and 8-rings composed of sp^2 and sp^3-hybridized carbon atoms known as C_{568} with layer thickness (d) = 1.0 Å, as shown in Fig. 1 (a). It is made up of three and four coordinated carbon atoms, which are represented by C1 (sp^2) and C2 (sp^3). C_{568} has a space group of *P-4m2* (115) with tetragonal lattice. The optimized lattice parameters of a square unit cell of C_{568} consisting of 13 carbon atoms are $a = b = 5.725$ Å with carbon-carbon (C–C) bond lengths in the range of 1.403 Å, 1.446 Å, 1.414 Å, and 1.56 Å, respectively.

From total energy calculations, C_{568} with energy −8.83 eV/atom is more stable compared to the pentagraphene [15] and tetrahexcarbon [17] sheets, which are also composed of (sp^2–sp^3)-hybridized carbon atoms (Fig. 1(b)). Moreover, C_{568} is energetically more stable compared to well-known 2D allotropes of carbon, such as planar T-graphene [41], α-graphyne [42], β-graphyne [42], δ-graphyne [42], ph-graphene [42], and graphdiyne [41,43]. We further calculated the formation energy of C_{568} and compared with the other 2D cabon allotropes using the expression [44]:

$$E_f = \frac{E_{2D}}{N_{2D}} - \frac{E_{3D}}{N_{3D}} \quad (1)$$

Here, E_{2D} and E_{3D} represent the energies of the two-dimensional and the respective bulk materials whereas N_{2D}, N_{3D} corresponds to the number of the atoms in the respective unit cells. A low formation energy indicates the ease of the fabrication of the materials from their bulk counterparts. The 2D materials possessing formation energy within 200 meV/atom have been extracted as free standing or suspended monolayers such as graphene, MoS_2, SnS_2, BN, SnSe, WS_2 [44]. On the other hand, 2D materials having E_f higher than 200 meV/atom are unlikely to synthesize and require a suitable stabilizing substrate [44]. Here, we calculated the formation energy of C_{568} and compared with the other 2D cabon allotropes such as graphdiyne [45,46], graphyne [47], pentahexoctite (PHO) [14] and pentagraphene [15]. Interestingly, the graphdyine with E_f 823 meV/atom has been synthesized experimentally on the copper surface [45]. The E_f of graphyne, graphdyine, PHO, pentagraphene, and C_{568} are 675 meV/atom, 823 meV/atom, 423 meV/atom, 968 meV/atom, 462 meV/atom, respectively. The E_f of C_{568} exceeds 200 meV/atom. However, it is significantly low compared to experimentally synthesized 2D materials such as graphdyine, silicene indicating the possibility of realization of C_{568}. We also performed the phonon calculations to test the dynamic stability of C_{568}-sheet. From Fig. 2(a), the absence of the negative phonon frequency in the phonon spectra further confirms its dynamic stability.

We further study the effect of equi-biaxial strain on the dynamical stability of C_{568} and compared with the graphene. By applying equi-biaxial tensile strain on C_{568}, we found it stable up to 11.2% strain, as shown in Fig. 2(b). As applied strain is increased up to 11.3%, C_{568} becomes unstable due to the appearance of negative phonon modes in the phonon dispersion curve (Supplementary Information, S1(a)). Similarly, we applied equi-biaxial strain to graphene and noticed that graphene remains dynamical stable up to 16.1% (Fig. 2(c)). However, graphene shows dynamic instability at 16.15% as negative phonon modes appear in the phonon spectra (Supplementary Information, S1(b)). This comparable strain limit of C_{568} to graphene before the appearance of dynamical instabilities indicate its suitability for nano-mechanical applications [48].

We further study the variation of stress with the equi-biaxial strain of C_{568} and compared with the pentahexoctite (PHO) and pentagraphene, Fig. 2(f). In the case of C_{568}, the strain

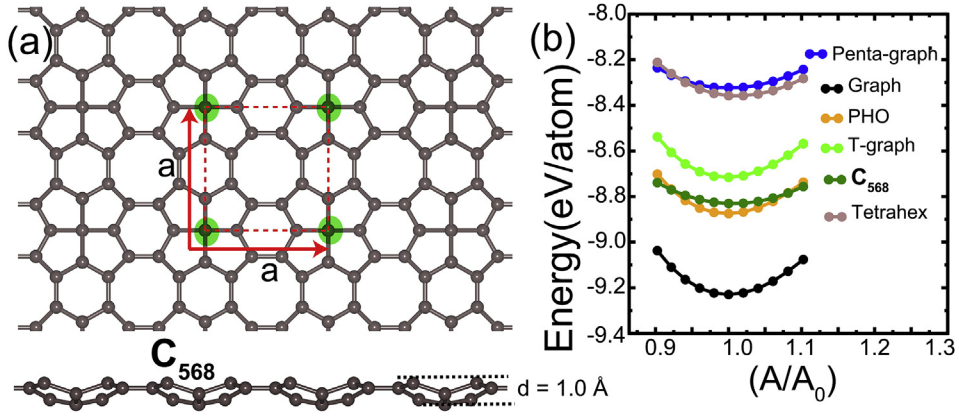


Fig. 1. (a) C_{568} sheet (top and side-views) with square unit-cell marked in red dotted lines, where *a* and *d* represent lattice vector (marked with solid red-arrow) and the total thickness of this sheet. (b) Energetics of C_{568} and its comparison with other 2D carbon allotropes such as Penta-graph (pentagraphene), Graph (graphene), PHO (pentahexoctite), T-graph (T-graphene) and Tetrahex (tetrahexcarbon), respectively. Here, the green-colored atoms represent C2 (sp^3). (A colour version of this figure can be viewed online.)

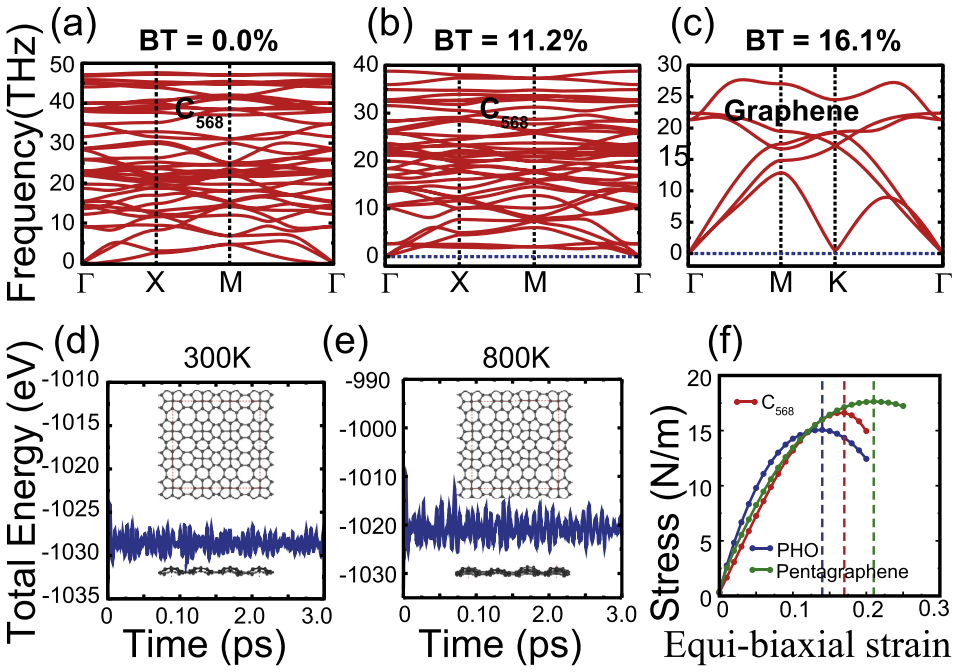


Fig. 2. (a) C_{568} phonon dispersion plot confirming its dynamical stability due to the absence of the negative phonon frequency. (b–c) Comparison of phonon spectra at extreme equi-biaxial tensile strain (11.2%) for C_{568} and graphene (16.1%) before dynamic instabilities occur, respectively. (d–e) Total energy variations and atomic configurations of 3 × 3 × 1 supercell (top and side views) of C_{568} at the end of the 3 ps AIMD simulations (inset) at room temperature (300 K) and 800 K, respectively. (f) Stress-strain curves as a function of equi-biaxial tensile strain of C_{568}, pentahexoctite (PHO), and pentagraphene 2D sheets. The red, blue, and green vertical dotted lines represent the strain values corresponding to maximum stress (top of curves). (A colour version of this figure can be viewed online.)

corresponding to the maximum stress is 17%, which is less compared to pentagraphene (21%) and higher than PHO (14%). From stress-strain curves, we also noticed the maximum stress value for each material beyond which a material shows mechanical failure, known as the ultimate tensile strength (UTS) of a material. For pentagraphene, UTS was found to be the highest of 17.63 N/m, whereas C_{568} shows UTS of 16.61 N/m. However, UTS of C_{568} is found higher than the pentahexoctite (PHO) sheet. Also, C_{568} shows phonon instability before reaching the maximum stress level, which has also been studied in the case of graphene [49]. Next, we investigate the effect of in-plane strain on the change in energy of C_{568} to determine its mechanical stability, which is very important for any nanomaterial from the application perspective. For two dimensional system, the elastic strain energy per unit area using standard Voigt notation [50] (i.e., 1-xx, 2-yy, and 6-xy) can be expressed as

$$U(\varepsilon) = \frac{1}{2}C_{11}\varepsilon_{xx}^2 + \frac{1}{2}C_{22}\varepsilon_{yy}^2 + C_{12}\varepsilon_{xx}\varepsilon_{yy} + 2C_{66}\varepsilon_{xy}^2 \quad (2)$$

where C_{11}, C_{22}, C_{12}, and C_{66} are elastic constants for 2D-system and can be obtained by fitting the energy curves related to the strain (uniaxial and biaxial). C_{568} has tetragonal symmetry, thus, $C_{11} = C_{22}$. Under the application of uniaxial strain, ($\varepsilon_{yy} = 0$), $U(\varepsilon) = \frac{1}{2}C_{11}\varepsilon_{xx}^2$, whereas, in the case of equi-biaxial strain ($\varepsilon_{xx} = \varepsilon_{yy}$), $U(\varepsilon) = (C_{11} + C_{12})\varepsilon_{xx}^2$. Elastic constants can be obtained by the parabolic fitting of the uniaxial and equi-biaxial strain curves. For uniaxial strain, C_{11} = 210.52 N/m, whereas for eqi-biaxial, C_{11}+C_{12} = 211.47 N/m, therefore, C_{12} = 0.95 N/m. Thus, C_{11}>|C_{12}| and the calculated C_{66} is positive, which further confirms its mechanical stability. Elastic constants can also be used to study the mechanical properties such as in-plane Young's modulus (E) and poisson's ratio (ν). The in-plane Youngs modulus calculated by $E = (C_{11}^2 - C_{12}^2)/C_{11}$, is found to be 210.52 N/m for C_{568}, which is lower than the graphene but comparable to the pentagraphene [15].

Unlike pentagraphene, the poisson's ratio (PR) as calculated by $\nu = C_{12}/C_{11}$ is found to be positive (ν = 0.0045) for C_{568}. For further confirmation of this nearly-zero PR, we calculated the lateral response in the y-direction by applying tensile strain along x-direction. Here, we applied the tensile strain (ε_{xx}) of 4%, 6%, and 8% along the x-direction and noticed that the lattice constant along y-direction remains unchanged in all three cases, as shown in Supplementary Information, S1(c). This further confirms the nearly-zero PR of C_{568}. It has been found that the materials like single-wall nanotube aerogel [51] and spongy graphene material [52] also show nearly-zero PR. In addtion, we test the theoretical cohesive strength limit based on the ratio of Young's modulus (E) and ultimate tensile strength (UTS) (the highest point of the stress-strain curve beyond which mechanical failure occurs) of C_{568} sheet [53]. For C_{568} (E = 210.52 N/m; UTS = 16.61 in N/m), E/UTS ratio was found to be 12.67, which is below the Griffith theoretical limit (UTS ≈ E/9) [53].

To study the effect of temperature on C_{568}, we employed AIMD-based simulations using canonical ensemble [54] at 300 K and 800 K for 3 ps (time step of 1 fs) in 3 × 3 × 1 supercell, as shown in Fig. 2(d and e)). Thereafter, we study the variation in total thickness (d) and the average bond length between carbon atoms in a 3 × 3 × 1 supercell of C_{568} as a function of time at 300 K and 800 K, as shown in Supplementary Information S2(b-d). The average total thickness (d_{avg}) of C_{568} at 800 K increases up to 1.73 Å compared to 1.62 Å at 300 K (Supplementary Information S2(b)). Also, the overall increase in the average bond lengths are noticed at 800 K (red solid curve) compared to 300 K (blue solid curve) between (C1–C1, C2–C1) (Supplementary Information S2(c)) and C2–C1 (Supplementary Information S2(d)). Beyond 800 K, the C–C bond lengths increase the experimentally observed C–C bond length of 1.704 Å [55]. As we know that C_{568} is composed of both 3- and 4-coordinated carbon atoms with the relatively larger C1–C2 bond

lengths, which further weaken the π-bonding compared to the flat sp^2-hybridized sheets. As a consequence, buckling increases, which is also found to increase significantly at 800 K compared to 300 K (Supplementary Information S2(b)). However, no significant structural reconstruction was observed indicating its thermal stability.

Moreover, it is well-known that the presence of defects plays an important role in influencing the structural stability of the materials. In 2D materials, especially vacancy defects are commonly observed during the epitaxial growth, at grain boundaries, or any place after heat treatment [56–58]. Here, we explored the effect of vacancy defects on C_{568} by calculating the vacancy formation energy using $E_{vac}=(E_{C+v} + nE_P)-E_C$ where, E_{C+v} and E_C are the total energies of C_{568} with and without vacancy, respectively. E_P is the energy per atom of bulk carbon (diamond) and n is the number of missing atom with a vacancy in C_{568}. In this study, we completely remove a carbon atom from its original position and create a single vacancy defect in a 2 × 2 × 1 supercell of C_{568} at each three inequivalent carbon atoms site (marked in blue circles) one by one, as shown in Supplementary Information S3(a-d). For these three inequivalent atomic sites, we have shown here three different atomic configurations represented by SV1, SV2, and SV3, respectively (Supplementary Information S3(b-d)). These atomic configurations (SV1, SV2, and SV3) are relaxed and shown with their top and side views (Supplementary Information S3 (b-d)). For SV1, the vacancy formation energy was found to be 7.7 eV with the total magnetic moments of 2.0 μ_B. Unlike SV1, both SV2, and SV3 are non-magnetic with formation energies of 2.1 eV and 6.11 eV, respectively. These three configurations possess high vacancy formation energy (Supplementary Information S3 (b-d)). Owing to the high formation energy of these configurations, the possibility of the formation of vacancy defects is low in C_{568}.

We further study the electronic properties of C_{568}. From band structure calculations, the conduction band minimum (CBM) and valence band maximum (VBM) located at Γ and M-symmetric points with the band gap of 1.13 eV (HSE06) making C_{568} an indirect band gap semiconductor, as shown in Fig. 3. As we know that C_{568} is composed of both C1 and C2 carbon atoms, the angular momentum resolved density of states (LDOS) analysis show that the band-edge positions (CBM and VBM) mainly originate from C1-p_z orbital (Fig. 3) as also reflected in the charge densities lobes (yellow color) around the C1-atoms (Fig. 3).

Furthermore, we compare the above mentioned properties of C_{568} with theoretically predicted and experimentally synthesized 2D sheets such has graphene [1,3], graphyne [47], graphdiyne [45,46], phagraphene [16], pentahexoctite [14], tetrahexcarbon [17], pentagraphene [15], as given in Table 1. Among these sheet, graphene holds the lowest formation energy (E_f) along with the incredible in-plane stiffness and thus, has been extracted as free standing with its ultimate strength. After graphene, phagraphene also shows low E_f with significantly high in-plane stiffness compared to C_{568}. Similarly, Pentahexoctite, a metallic carbon sheet composed of 5-6-8 rings of carbon atoms like C_{568} possesses low E_f as well as high in-plane stiffness compared to C_{568}. However, the zero-gap and metallic nature of graphene, phagraphene, and pentahexoctite (PHO) sheets restrict their application in FETS [22–24].

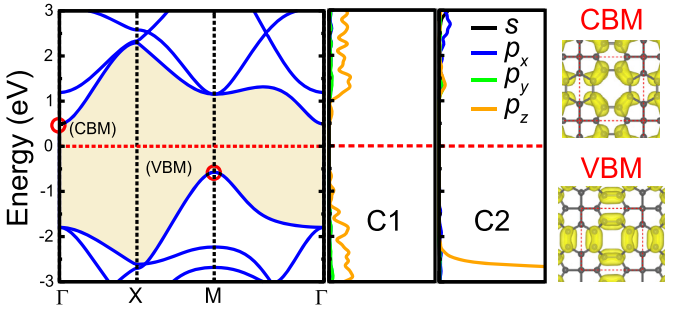


Fig. 3. Electronic band structure of C_{568} calculated using (HSE06) with the LDOS for C1 (sp^2) and C2 (sp^3) atoms. The band decomposed charge densities (yellow-colored lobes) at band edge positions (CBM and VBM). Here, red dotted line and red circles represent the Fermi energy level and the band edge positions, respectively. The isosurface is set to (0.02 eV/Å^3). (A colour version of this figure can be viewed online.)

In addition to graphene, graphdiyne (sp–sp^2-hybridized sheet) a substructure of graphyne (sp–sp^2-hybridized sheet) has also been realized experimentally on the copper surface [45]. Unlike graphene, graphdiyne and graphyne are semiconductors with the direct band gap (HSE06) of 1.12 eV [46] and 0.96 eV [47], respectively. However, the linear atomic chains of graphdiyne and graphyne further reduce their in-plane stiffness and found to be lowest among these 2D sheets (Table 1). Moreover, pentagraphene and tetrahexcarbon composed of sp^2–sp^3-hybridized carbon atoms also show interesting electronic and optoelectronic properties with in-plane stiffness higher than C_{568}. However, the E_f is found to be highest for these sheets (Table 1). It is known that the 2D materials having E_f higher than 200 meV/atom are unlikely to synthesize and require a suitable stabilizing substrate, such as graphdiyne, silicene, AlN, and ZnO [44,45]. Thus, we expect that C_{568} with its reasonably low formation energy (462 meV/atom) among the semiconducting sheets (Table 1) along with the in-plane stiffness of 210.52 N/m could be one of the 2D material for the nanoelectronic applications.

As we know that the electronic transport properties are mainly governed by the electronic mobility which is strongly influenced by the charge carriers effective masses. In addition to that carrier mobilities also depend on the scattering mechanism of the charge carriers. In this study, we assume the electron acoustic phonon scattering mechanism as a key factor for restricting the charge carrier mobility. For two-dimensional isotropic materials, the charge carrier mobility (μ) is defined by Ref. [28]:

$$\mu = \frac{e\hbar^3 C_s}{k_B T m_x^* \sqrt{\left(m_x^*\right)\left(m_y^*\right)}\left(E_1^{i2}\right)} \tag{3}$$

where, m_x^* (m_y^*), T, C_s, and E_1 are the effective masses of the charge carriers, temperature, C_s in-plane stiffness, and deformation potential constant of band edge positions (VBM or CBM) along transport directions. The effective masses are calculated in the vicinity of Γ-point by fitting the dispersive bands to $m_x^* = \frac{\hbar^2}{\partial^2 E / \partial k^2}$ along the x-direction, where $m_x^* = m_y^*$ for the tetragonal lattice. The deformation potential, is defined by $E_1^i = \frac{\triangle V_i}{\triangle a / a_0}$, where, $\triangle V_i$ is defined as the variation in ith energy band under compression or expansion (step of 0.5%). Here, a and a_0 are lattice parameters under strain and at equilibrium condition. The deformation potential constant (E_1) is obtained from the linear fits to the lattice dilation and the respective band edge positions (Supplementary Information, S6(a-b)). The transport properties results for C_{568} are summarized in Table 2. For C_{568}, the charge carrier mobilities along the x-direction are calculated at 300 K. The electron mobility along x-direction is ~1.366 × 10^4 $cm^2V^{-1}s^{-1}$ whereas, hole mobility is ~0.11 × 10^3 $cm^2V^{-1}s^{-1}$, as given in Table 2. The high mobility of electron compared to the hole is mainly due to the large differences in effective masses between electron and hole compared to the difference in their respective deformation potential.

In addition to C_{568}, we also studied structural, electronic and transport properties of its stacked configurations. C_{568} can be stacked in three different configurations such as Type-1, Type-2, and Type-3 consisting of 26 carbon atoms, as shown in Fig. 4(a–c).

Table 1
The formation energy (eV/atom), carbon atoms density (atom/Å^2), in-plane stiffness, and electronic properties of 2D sheets such as graphene, graphyne, graphdiyne, phagraphene, pentahexoctite (PHO), tetrahexcarbon, pentagraphene. The formation energies of these carbon sheets are calculated with respect to bulk carbon (graphite). The "S", "M", "D" represents semiconductor, metal and Dirac semimetal.

2D sheets (Hybridization)	Carbon rings	Carbon density (atom/Å^2)	Formation energy (meV/atom)	In-plane stiffness (N/m)	Electronic properties
graphene [1,3] (sp^2)	6	0.379	60	345 ± 50	D
phagraphene [59,60] (sp^2)	5 + 6+7	0.371	260	313.21	D
PHO [14] (sp^2)	5 + 6+8	0.362	423	299.4	M
C_{568} (sp^2-sp^3)	5 + 6+8	0.397	462	210.52	S
graphyne [47] (sp-sp^2)	6 + 12	0.292	675	165.96	S
graphdiyne [45,46] (sp-sp^2)	6 + 18	0.233	823	121	S
tetrahexcarbon [17] (sp^2-sp^3)	4 + 6	0.452	924	287.9	S
pentagraphene [15] (sp^2-sp^3)	5	0.452	968	265	S

Table 2
Calculated effective masses (m_x^*), in-plane stiffness (C_x in (Nm^{-1})), deformation potential constants (E_{1x} in (eV)), electrons and hole mobilities (μ_x in (10^3 cm$^2V^{-1}s^{-1}$)) for C_{568}, Type-1, and Type-2. Carrier types (e) and (h) represent electrons and holes, respectively. Here, BP stands for black phosphorus.

Layer number	System	Carrier types	m_x^*/m_0	E_{1x}	C_x	μ_x (BP) [28]
1	C_{568}	e	0.17	3.37	210.52	13.66 (**1.1**–**1.14**)
1	C_{568}	h	0.64	10.05	210.52	0.11 (**0.64**–**0.7**)
2	Type-1	e	0.145	4.23	430.38	24.35 (**0.6**)
2	Type-1	h	0.61	12.36	430.38	0.16 (**2.6**–**2.8**)
2	Type-2	e	0.15	7.81	439.91	6.82
2	Type-2	h	0.67	6.51	439.91	0.49

Type-1 configuration is obtained by stacking two C_{568} layers on top of each other, whereas Type-2 is obtained by sliding the top layer of Type-1 followed by the optimization and relaxations. For Type-1, lattice parameters are $a = b = 5.725$ Å with the C–C bond lengths similar to that of monolayer C_{568} with the interlayer distance (d_1) = 2.8 Å, as shown in Fig. 4(a). Like Type-1, the optimized lattice parameters of Type-2 configuration are similar with interlayer distance (d_2) = 3.04 Å. The C–C bond lengths are 1.405 Å, 1.45 Å, 1.42 Å, and 1.56 Å, respectively (Fig. 4(b)). Unlike Type-1 and Type-2, Type-3 configuration is obtained by flipping the top layer of Type-1 followed by the optimization and relaxation. Type-3 has a rectangular unit cell with the optimized lattice parameters $a = 5.606$ Å; and $b = 5.72$ Å and C–C bond lengths 1.35 Å, 1.415 Å, 1.44 Å, 1.53 Å, and 1.604 Å, respectively, as shown in Fig. 4(c). Energetically, Type-1 (−8.88 eV/atom) and Type-2 (−8.884 eV/

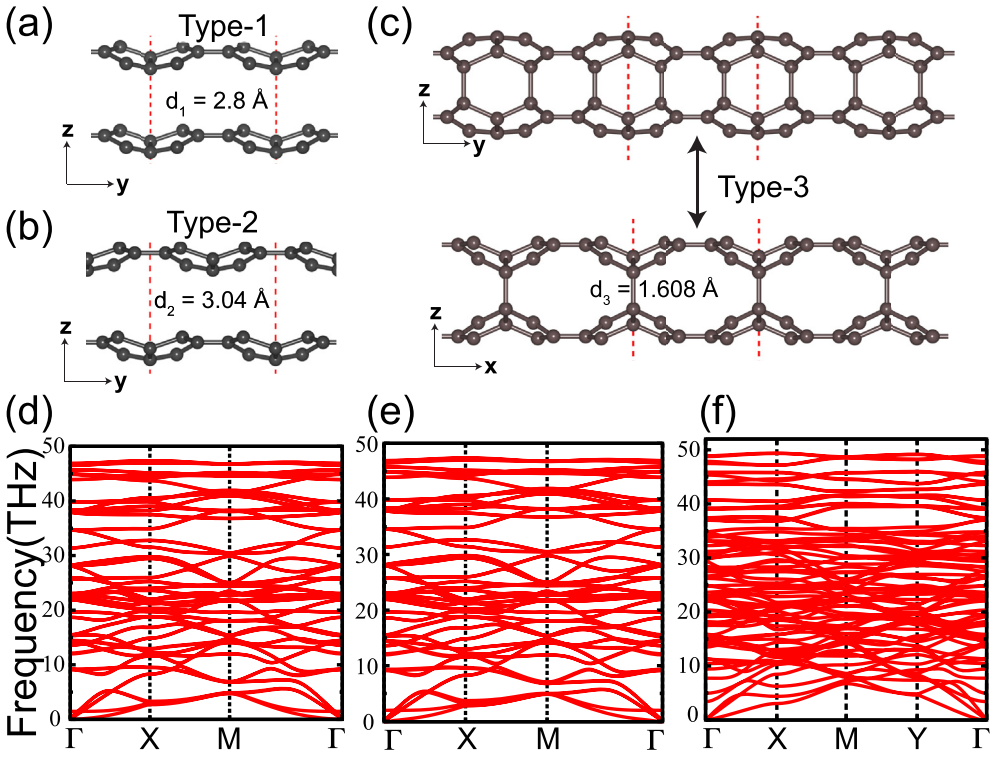


Fig. 4. (a–b) Bilayers (Type-1 and Type-2) of C_{568} having side-views along *zy*-plane with interlayer distances $d_1 = 2.8$ Å and $d_2 = 3.04$ Å, respectively. (c) Type-3, shown with *zx* and *zy*-planes (side-views) with intra-layer distance $d_3 = 1.608$ Å. (d–f) Phonon dispersion plots of Type-1, Type-2 and Type-3 bilayers confirming their dynamical stabilities. (A colour version of this figure can be viewed online.)

atom) are more stable compared to Type-3 (−8.831 eV/atom). These configurations also show dynamic stabilities as negative phonon modes are absent in the phonon spectra, as shown in Fig. 4(d–f). In addition to C_{568}, we also test the theoretical cohesive strength limit based on E/UTS ration for its bilayers (Type-1, Type-2, Type-3). For Type-1 (E = 424.9 N/m; UTS = 33.3 N/m), Type-2 (E = 421.6 N/m; UTS = 33.3 N/m), and Type-3 (E = 347.52 N/m; UTS = 34.31 N/m) structures E/UTS ratio were 12.76, 12.66, and 10.13 (Supplementary Information S2(a)), which are found below the Griffith theoretical limit (UTS ≈ E/9) [53].

We also study the effect of temperature on C_{568} bilayer (Type-1) in addition to the weak vdW interactions by employing AIMD-based simulations at 300 K, and 800 K for 3 ps (time step of 1 fs) in 3 × 3 × 1 supercell. Recently, it has been shown that the thermally-induced ripples play an important role in influencing the interaction strength between graphene and copper substrate [61]. Here, we calculated the interlayer distance (d_1) as a function of time at 300 K and 800 K (Supplementary Information S4(a)), respectively. The overall reduction in d_1 was observed with an average interlayer distance (d_{1avg}) of 2.46 Å (300 K) and 2.32 Å (800 K) (Supplementary Information S4(a)). During the simulations, we observed an atomic configuration (inset) with the lowest interlayer distance of 1.59 Å at 2.72 ps (at 800 K) having a C–C bond length (yellow dotted lines) of 2.51 Å between the carbon atoms of top and bottom layers (Supplementary Information S4(a)). This C–C bond length of 2.51 Å is even larger than the experimentally known longest C–C bond length of 1.77 Å [15]. Moreover, the interlayer distance (d_1) increased up to 2.57 Å (800 K) with C–C = 2.99 Å and 2.30 Å (300 K) with C–C = 3.1 Å at the end of 3ps AIMD simulations, as shown in Supplementary Information S4(a). These results show that no bond formation was observed during the simulation at temperature 300 K and 800 K indicating the existence of weaker interlayer interactions in bilayer C_{568}.

We further study the electronic properties of C_{568} bilayers. From band structure calculation (Fig. 5(a)), the band edge positions (VBM and CBM) of Type-1 lies at Γ-and M-points with the band gap of 0.814 eV (HSE06) making it an indirect band gap semiconductor. Similarly, Type-2 has band gap of 0.96 eV (HSE06) at Γ- and close to M-point making it an indirect band gap semiconductor, as shown in Fig. 5(b). The orbital contribution towards the band edge positions in both Type-1 and Type-2 are mainly coming from C1-p_z orbitals as shown in LDOS analysis (Fig. 5(a and b)). This is further confirmed from the charge density plots (yellow color lobes) at VBM and CBM (Supplementary Information, S5(c-d)).

In addition to Type-1 and Type-2 bilayers, we also explored the electronic properties of Type-3 bilayer. In Type-3, layers are covalently bonded with the reduced interlayer distance (d_3 = 1.608 Å) rather than via weak vdW interactions as in the case of Type-1 and Type-2, which also leads to the change in lattice parameters. As a consequence, the strong interactions between off-plane orbitals (C1-p_z and C2-p_z) lead energy bands to cross the Fermi level and thus flipping induced semiconductor to metal transition is realized (Fig. 5(c)). Interestingly, Type-3 also forms a 12-carbon atoms ring (dodecagonal) hole normal to the *zx*-plane, unlike other cases. The presence of the dodecagonal hole (Fig. 5(c)) in Type-3 along with its thermal stability at 300 K and 800 K (Supplementary Information, S5(a-b)) might be useful as a porous 2D membrane for selective diffusion of atoms or molecules, whereas graphene is found to be impermeable [62].

We further investigate the effect of applied strain on the electronic properties of semiconducting Type-1 and Type-2 configurations. Amazingly, 2D materials are known to perform better under strain conditions as they can sustain larger mechanical strain compared to their bulk counterparts [19,63]. This strain can be applied to open the band gap of semi-metallic graphene [64,65],

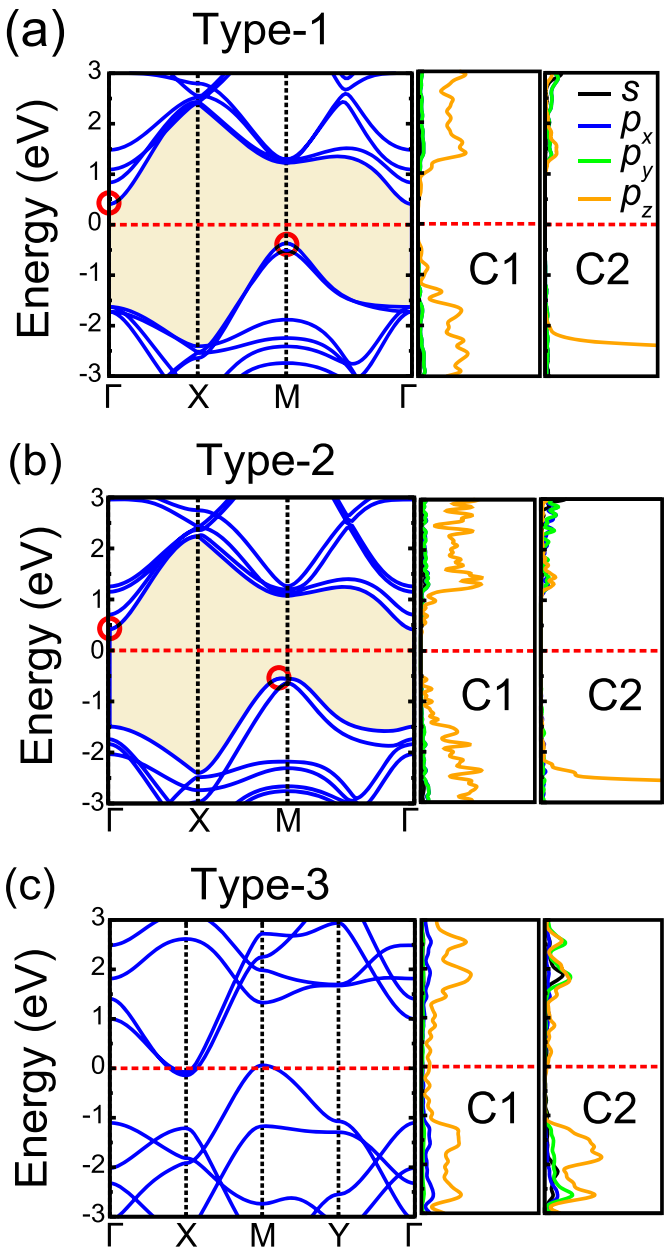


Fig. 5. (a–c) Electronic band structures of Type-1, Type-2, and Type-3 calculated using (HSE06) with LDOS for C1 and C2 atoms. Here, red circles correspond to CBM and VBM. The red dotted line represents Fermi energy levels, respectively. (A colour version of this figure can be viewed online.)

the transition from indirect to direct or to realize the semiconductor to metal transitions [66–68]. Overall, the strain can be applied to tune mechanical, optical, thermal, piezo, and photocatalytic properties [69–72].

Here, we also study the change in band gap as a function of applied normal compressive (NC) strain for Type-1 and Type-2 using DFT-PBE. The NC strain is more feasible from the practical aspects compared to the applied in-plane strains due to the weak vdW interactions between the layers compared the strong intralayer bonding between the atoms. NC strain (ε_{NC}) is defined as $\varepsilon_{NC} = \frac{z-z_0}{z_0}$, where z and z_0 are the interlayer thickness under applied strain and zero strain. For unstrained case, band gap of Type-1 and Type-2 are 0.416 eV and 0.533 eV, respectively. As we increase NC strain up to 6%, 12%, and 19%, the decrease in band gap

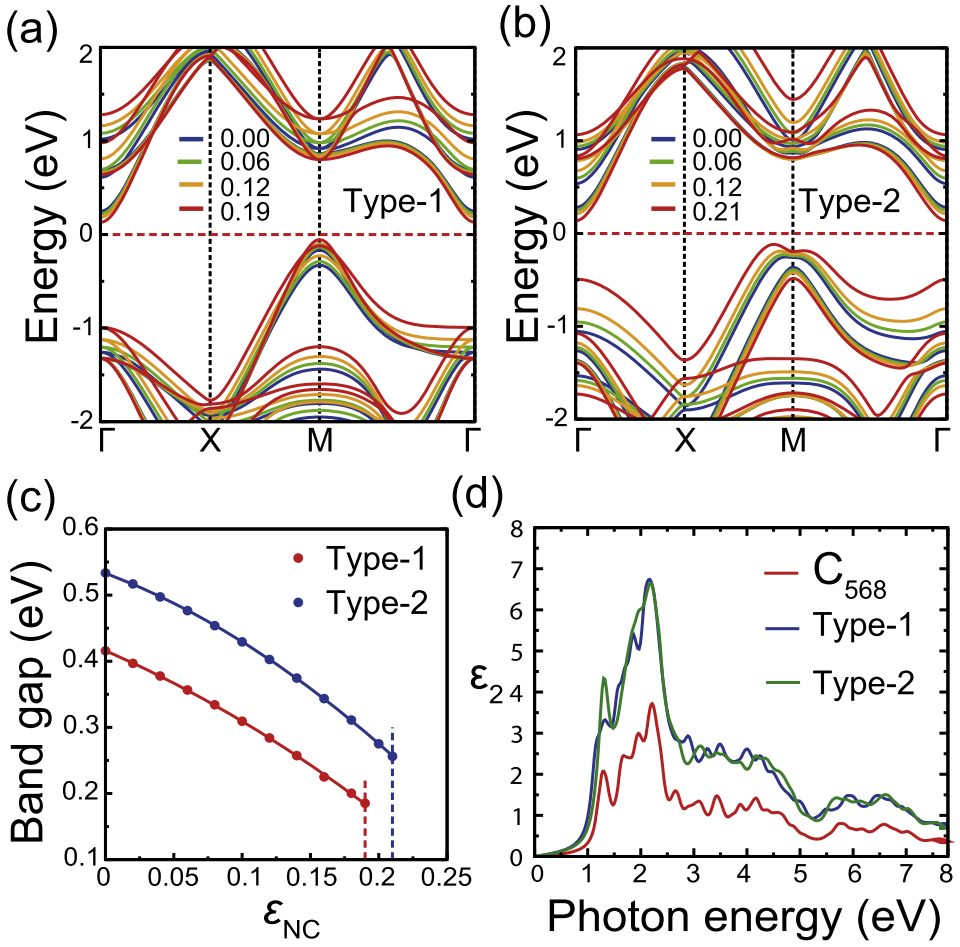


Fig. 6. (a–b) Electronic band structures of Type-1 and Type-2 under applied normal compressive (NC) strains (ε_{NC} = 0.0,0.06,0.12,0.19,0.21). (c) Band gap variation in Type-1 and Type-2 as a function of normal compressive (NC) strain (ε_{NC}). Red and blue vertical-dotted lines represent the NC strain limits (0.19 and 0.21) where dynamic instabilities occur. (d) The imaginary part of dielectric function (ε_2) as a function of photon energy of C_{568}, Type-1, and Type-2. (A colour version of this figure can be viewed online.)

is observed up to 19% of applied strain, as shown in Fig. 6(a–c). At extreme NC strain of 19% and 21%, band gap reduced to 0.185 eV (Type-1) and 0.256 eV (Type-2), respectively. However, the band gap remains indirect throughout the applied NC strain (19%) indicating its insensitivity towards NC strain. Beyond 19% and 21%, structures show dynamic instabilities (marked by red and blue vertical-dotted lines), as shown in Fig. 6(c). Such a tunability of band gap of C_{568} bilayers could be beneficial to the experimentalists for exploring their possible applications. For Type-1, the change in band gap as a function of NC strain can be modeled by the equation:

$$E_g = 0.41595 - 0.89674\varepsilon_{NC} - 1.7037\varepsilon_{NC}^2 \quad (4)$$

and for Type-2 by:

$$E_g = 0.53322 - 0.78406\varepsilon_{NC} - 2.5275\varepsilon_{NC}^2 \quad (5)$$

By substituting the desired ε_{NC} value in the above equations, we can obtain the different band gap (E_g) values. This tunability of C_{568} bilayers at the electronic level might find possible applications in flexible and strain tunable 2D electronic devices.

Similar to C_{568}, we also study the transport properties of bilayers (Type-1 and Type-2). The transport properties results for Type-1 and Type-2 are also summarized in Table 2. The deformation potential constants calculated for these structures are given in (Supplementary Information, S6(c-f)). Like C_{568}, bilayers also show extraordinary electronic mobilities at room temperature along the *x*-direction. The predicted electronic mobility for Type-1 is ~2.435 × 10^4 $cm^2V^{-1}s^{-1}$, whereas Type-2 have ~6.82 × 10^3 $cm^2V^{-1}s^{-1}$ along *x*-direction. The hole mobility for Type-1 is found to be ~0.16 × 10^3 $cm^2V^{-1}s^{-1}$, whereas Type-2 shows ~0.49 × 10^3 $cm^2V^{-1}s^{-1}$, which are smaller than the electronic mobilities along the *x*-direction. The increase in electronic mobilities in the case of Type-1 compared to Type-2 and C_{568} is due to the lower effective masses of electrons as well as deformation potential, as given in Table 2. The predicted electron mobility for C_{568} (~1.366 × 10^4 $cm^2V^{-1}s^{-1}$) and Type-1 (~2.435 × 10^4 $cm^2V^{-1}s^{-1}$) are higher than the black phosphorus monolayer (~1.1 × 10^3 $cm^2V^{-1}s^{-1}$) [28] monolayers, whereas comparable to the Tetrahexcarbon (~1.46 × 10^4 $cm^2V^{-1}s^{-1}$) [17]. Thus, we believe that C_{568} and its bilayer configurations might find potential application in FETs [28].

Moreover, we also explored the optoelectronic properties of C_{568} and its bilayers. The dielectric function ($\varepsilon(\omega)$) is calculated by using the independent particle approximation as implemented in VASP [73]. From Fig. 6(d), the imaginary part ($\varepsilon_2(\omega)$) is plotted as a function of photon energy for C_{568}, Type-1, and Type-2 structures. For bilayers, the absorption spectra is red-shifted compared to the C_{568} due to the reduction in band gap of Type-1 (0.814 eV) and Type-2 (0.96 eV) compared to C_{568} (1.13 eV). The origin of absorption peaks can be attributed to the interband transitions between the *p* states of C1–C1 and C1–C2 atoms as shown in density

of states calculations of these materials. The absorption spectra of monolayer (C_{568}), bilayers (Type-1 and Type-2) are similar. However, increase in absorption is observed in bilayers compared to C_{568}, which is commonly observed trend as the number of layers increases. The optical absorption edges are found to start approximately from the band gap of C_{568}, Type-1, and Type-2 indicating the tuning absorption edge by changing of number of layers.

4. Conclusion

In conclusion, we designed a new two-dimensional allotrope of carbon C_{568} by the combination of multi-start local search approach and atomistic simulation using Tersoff potential, which is further optimized using first-principles calculations. C_{568} is semiconducting with an indirect band gap of 1.13 eV (HSE06) at Γ-and M-symmetric points. This sheet exhibits energetic, dynamic, mechanical and thermal stabilities and also can withstand temperature up to 800 K. C_{568} shows its ideal strength comparable to pentagraphene. It shows in-plane Young's modulus comparable to pentagraphene [15] with positive poisson's ratio. Interestingly, C_{568} exhibits electronic mobility of order ~10^4 $cm^2V^{-1}s^{-1}$ along the *x*-direction, which is higher than black phosphorus monolayer, and comparable to Tetrahexcarbon sheet.

In addition to C_{568}, its bilayers (Type-1 and Type-2) are also semiconductors with band gap of 0.814 eV and 0.96 eV, respectively. Along with the dynamic stabilities, bilayers (Type-1 and Type-2) also show extraordinary electronic mobility of order ~$10^3$$cm^2V^{-1}s^{-1}$ and ~10^4 $cm^2V^{-1}s^{-1}$. We also noticed that the band gap of Type-1 and Type-2 configurations remain indirect throughout the applied NC strain indicating its insensitivity towards applied NC strain. The possibility of the realization of C_{568} can be understood from its energetic, dynamical, mechanical, and thermal stabilities, which further indicates its future applications in nanoelectronics.

Author contribution statement

Dr. Hiroshi Mizuseki: worked on method development, reviewing and supervision

Dr. Babu Ram: worked conceptualization of problem, draft preparation, reviewing and editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the Korea Institute of Science and Technology (Grant No.2E28000) for getting financial support to accomplish this work. The authors also acknowledge the support from HPCI system (Project ID: hp180036, hp180125) and the inter-university cooperative research program of the Institute for Materials Research, Tohoku University (Proposal No. 18S0001), Japan.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbon.2019.11.062.

References

[1] K.S. Novoselov, A.K. Geim, S.V. Morozov, D. Jiang, Y. Zhang, S.V. Dubonos, I.V. Grigorieva, A.A. Firsov, Electric field effect in atomically thin carbon films, Science 306 (2004) 666–669.
[2] K.S. Novoselov, A.K. Geim, S.V. Morozov, D. Jiang, M.I. Katsnelson, I.V. Grigorieva, S.V. Dubonos, A.A. Firsov, Two-dimensional gas of massless Dirac fermions in graphene, Nature 438 (7065) (2005) 197–200.
[3] A.K. Geim, K.S. Novoselov, The rise of graphene, Nat. Mater. 6 (3) (2007) 183–191.
[4] R.H. Baughman, H. Eckhardt, M. Kertesz, Structure-property predictions for new planar forms of carbon: layered phases containing sp^2 and *sp* atoms, J. Chem. Phys. 87 (11) (1987) 6687–6699.
[5] J. Zhou, K. Lv, Q. Wang, X.S. Chen, Q. Sun, P. Jena, Electronic structures and bonding of graphyne sheet and its BN analog, J. Chem. Phys. 134 (17) (2011) 174701.
[6] N. Narita, S. Nagai, S. Suzuki, K. Nakao, Optimized geometries and electronic structures of graphyne and its family, Phys. Rev. B 58 (1998) 11009–11014.
[7] G. Li, Y. Li, H. Liu, Y. Guo, Y. Li, D. Zhu, Architecture of graphdiyne nanoscale films, Chem. Commun. 46 (2010) 3256–3258.
[8] M. Long, L. Tang, D. Wang, Y. Li, Z. Shuai, Electronic structure and carrier mobility in graphdiyne sheet and nanoribbons: theoretical predictions, ACS Nano 5 (4) (2011) 2593–2600.
[9] Q. Song, B. Wang, K. Deng, X. Feng, M. Wagner, J.D. Gale, K. Mullen, L. Zhi, Graphenylene, a unique two-dimensional carbon network with nondelocalized cyclohexatriene units, J. Mater. Chem. C 1 (2013) 38–41.
[10] E. Konstantinova, S.O. Dantas, P.M.V.B. Barone, Electronic and elastic properties of two-dimensional carbon planes, Phys. Rev. B 74 (2006), 035417.
[11] T. Lange, V. Gramlich, W. Amrein, F. Diederich, M. Gross, C. Boudon, J.-P. Gisselbrecht, Hexakis(trimethylsilylethynyl)[3]radialene: a carbon-rich chromophore with unusual electronic properties, Angew. Chem. Int. Ed. 34 (7) (1995) 805–809.
[12] V.H. Crespi, L.X. Benedict, M.L. Cohen, S.G. Louie, Prediction of a pure-carbon planar covalent metal, Phys. Rev. B 53 (1996) R13303–R13305.
[13] H. Terrones, M. Terrones, E. Hernández, N. Grobert, J.-C. Charlier, P.M. Ajayan, New metallic allotropes of planar and tubular carbon, Phys. Rev. Lett. 84 (2000) 1716–1719.
[14] B.R. Sharma, A. Manjanath, A.K. Singh, pentahexoctite, A new two-dimensional allotrope of carbon, Sci. Rep. 4 (2014) 7164.
[15] S. Zhang, J. Zhou, Q. Wang, X. Chen, Y. Kawazoe, P. Jena, Penta-graphene: a new carbon allotrope, Proc. Natl. Acad. Sci. 112 (8) (2015) 2372–2377.
[16] Z. Wang, X.-F. Zhou, X. Zhang, Q. Zhu, H. Dong, M. Zhao, A.R. Oganov, Phagraphene: a low-energy graphene allotrope composed of 5–6–7 carbon rings with distorted Dirac cones, Nano Lett. 15 (2015) 6182–6186.
[17] B. Ram, H. Mizuseki, Tetrahexcarbon: a two-dimensional allotrope of carbon, Carbon 137 (2018) 266–273.
[18] Y. Liu, G. Wang, Q. Huang, L. Guo, X. Chen, Structural and electronic properties of *t* graphene: a two-dimensional carbon allotrope with tetrarings, Phys. Rev. Lett. 108 (2012) 225505.
[19] C. Lee, X. Wei, J.W. Kysar, J. Hone, Measurement of the elastic properties and intrinsic strength of monolayer graphene, Science 321 (5887) (2008) 385–388.
[20] Y. Zhu, S. Murali, W. Cai, X. Li, J.W. Suk, J.R. Potts, R.S. Ruoff, Graphene and graphene oxide: synthesis, properties, and applications, Adv. Mater. 22 (35) (2010) 3906–3924.
[21] F. Schwierz, Graphene transistors, Nat. Nanotechnol. 5 (2010) 487.
[22] M. Kim, N.S. Safron, E. Han, M.S. Arnold, P. Gopalan, Fabrication and characterization of large-area, semiconducting nanoperforated graphene materials, Nano Lett. 10 (4) (2010) 1125–1131.
[23] V. Hung Nguyen, M. Chung Nguyen, H.-V. Nguyen, P. Dollfus, Disorder effects on electronic bandgap and transport in graphene-nanomesh-based structures, J. Appl. Phys. 113 (1) (2013) 013702.
[24] Y. Pan, L. Zhang, L. Huang, L. Li, L. Meng, M. Gao, Q. Huan, X. Lin, Y. Wang, S. Du, H.-J. Freund, H.-J. Gao, Construction of 2d atomic crystals on transition metal surfaces: graphene, silicene, and hafnene, Small 10 (11) (2014) 2215–2225.
[25] B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, A. Kis, Single-layer MoS_2 transistors, Nat. Nanotechnol. 6 (2011) 147.
[26] W. Choi, N. Choudhary, G.H. Han, J. Park, D. Akinwande, Y.H. Lee, Recent development of two-dimensional transition metal dichalcogenides and their applications, Mater. Today 20 (3) (2017) 116–130.
[27] X.-L. Sheng, Q.-B. Yan, F. Ye, Q.-R. Zheng, G. Su, T-carbon: a novel carbon allotrope, Phys. Rev. Lett. 106 (2011) 155703.
[28] J. Qiao, X. Kong, Z.-X. Hu, F. Yang, W. Ji, High-mobility transport anisotropy and linear dichroism in few-layer black phosphorus, Nat. Commun. 5 (2014) 4475.
[29] J. Tersoff, Modeling solid-state chemistry: interatomic potentials for multicomponent systems, Phys. Rev. B 39 (1989) 5566–5568.
[30] Z.-Z. Li, J.-T. Wang, H. Mizuseki, C. Chen, Computational discovery of a new rhombohedral diamond phase, Phys. Rev. B 98 (2018) 094107.
[31] W. Kohn, L.J. Sham, Self-consistent equations including exchange and correlation effects, Phys. Rev. 140 (1965) A1133–A1138.
[32] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, Comput. Mater. Sci. 6 (1) (1996) 15–50.
[33] G. Kresse, J. Furthmüller, Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set, Phys. Rev. B 54 (1996) 11169–11186.

[34] P.E. Blöchl, Projector augmented-wave method, Phys. Rev. B 50 (1994) 17953–17979.
[35] G. Kresse, D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B 59 (1999) 1758–1775.
[36] J. Heyd, J.E. Peralta, G.E. Scuseria, R.L. Martin, Energy band gaps and lattice parameters evaluated with the heyd-scuseria-ernzerhof screened hybrid functional, J. Chem. Phys. 123 (17) (2005) 174101.
[37] B.G. Janesko, T.M. Henderson, G.E. Scuseria, Screened hybrid density functionals for solid-state chemistry and physics, Phys. Chem. Chem. Phys. 11 (2009) 443–454.
[38] J.K. Ellis, M.J. Lucero, G.E. Scuseria, The indirect to direct band gap transition in multilayered MoS_2 as predicted by screened hybrid density functional theory, Appl. Phys. Lett. 99 (26) (2011) 261908.
[39] H.J. Monkhorst, J.D. Pack, Special points for brillouin-zone integrations, Phys. Rev. B 13 (1976) 5188–5192.
[40] S. Grimme, Semiempirical gga-type density functional constructed with a long-range dispersion correction, J. Comput. Chem. 27 (15) (2006) 1787–1799.
[41] Y. Liu, G. Wang, Q. Huang, L. Guo, X. Chen, Structural and electronic properties of *t* graphene: a two-dimensional carbon allotrope with tetrarings, Phys. Rev. Lett. 108 (2012) 225505.
[42] X. Zhang, L. Wei, J. Tan, M. Zhao, Prediction of an ultrasoft graphene allotrope with Dirac cones, Carbon 105 (2016) 323–329.
[43] G. Li, Y. Li, H. Liu, Y. Guo, Y. Li, D. Zhu, Architecture of graphdiyne nanoscale films, Chem. Commun. 46 (2010) 3256–3258.
[44] A.K. Singh, K. Mathew, H.L. Zhuang, R.G. Hennig, Computational screening of 2d materials for photocatalysis, J. Phys. Chem. Lett. 6 (6) (2015) 1087–1098.
[45] G. Li, Y. Li, H. Liu, Y. Guo, Y. Li, D. Zhu, Architecture of graphdiyne nanoscale films, Chem. Commun. 46 (2010) 3256–3258.
[46] Y. Pei, Mechanical properties of graphdiyne sheet, Phys. B Condens. Matter 407 (22) (2012) 4436–4439.
[47] J. Kang, J. Li, F. Wu, S.-S. Li, J.-B. Xia, Elastic, electronic, and optical properties of two-dimensional graphyne sheet, J. Phys. Chem. C 115 (42) (2011) 20466–20470.
[48] T. Hallam, C.F. Moldovan, K. Gajewski, A.M. Ionescu, G.S. Duesberg, Large area suspended graphene for nano-mechanical devices, Phys. Status Solidi B 252 (11) (2015) 2429–2432.
[49] C.A. Marianetti, H.G. Yevick, Failure mechanisms of graphene under tension, Phys. Rev. Lett. 105 (2010) 245502.
[50] R.C. Andrew, R.E. Mapasha, A.M. Ukpong, N. Chetty, Mechanical properties of graphene and boronitrene, Phys. Rev. B 85 (2012) 125428.
[51] K.H. Kim, Y. Oh, M.F. Islam, Mechanical and thermal management characteristics of ultrahigh surface area single-walled carbon nanotube aerogels, Adv. Funct. Mater. 23 (3) (2013) 377–383.
[52] Y. Wu, N. Yi, L. Huang, T. Zhang, S. Fang, H. Chang, N. Li, J. Oh, J.A. Lee, M. Kozlov, A.C. Chipara, H. Terrones, P. Xiao, G. Long, Y. Huang, F. Zhang, L. Zhang, X. Lepro, C. Haines, M.D. Lima, N.P. Lopez, L.P. Rajukumar, A.L. Elias, S. Feng, S.J. Kim, N.T. Narayanan, P.M. Ajayan, M. Terrones, A. Aliev, P. Chu, Z. Zhang, R.H. Baughman, Y. Chen, Three-dimensionally bonded spongy graphene material with super compressive elasticity and near-zero poissonâ??s ratio, Nat. Commun. 6 (1) (2015) 6141.
[53] H. Sun, S. Mukherjee, Z. Shi, C.V. Singh, Elastomer-like deformation in high-Poisson's-ratio graphene allotropes may allow tensile strengths beyond theoretical cohesive strength limits, Carbon 143 (2019) 752–761.
[54] S. Nosé, A unified formulation of the constant temperature molecular dynamics methods, J. Chem. Phys. 81 (1) (1984) 511–519.
[55] P.R. Schreiner, L.V. Chernish, P.A. Gunchenko, E.Y. Tikhonchuk, H. Hausmann, M. Serafin, S. Schlecht, J.E.P. Dahl, R.M.K. Carlson, A.A. Fokin, Overcoming lability of extremely long alkane carbonâ??carbon bonds through dispersion forces, Nature 477 (2011) 308.
[56] V.O. Özçelik, S. Cahangirov, S. Ciraci, Epitaxial growth mechanisms of graphene and effects of substrates, Phys. Rev. B 85 (2012) 235456.
[57] H. Chen, W. Zhu, Z. Zhang, Contrasting behavior of carbon nucleation in the initial stages of graphene epitaxial growth on stepped metal surfaces, Phys. Rev. Lett. 104 (2010) 186101.
[58] P.Y. Huang, C.S. Ruiz-Vargas, A.M. van der Zande, W.S. Whitney, M.P. Levendorf, J.W. Kevek, S. Garg, J.S. Alden, C.J. Hustedt, Y. Zhu, J. Park, P.L. McEuen, D.A. Muller, Grains and grain boundaries in single-layer graphene atomic patchwork quilts, Nature 469 (7330) (2011) 389–392.
[59] Z. Wang, X.-F. Zhou, X. Zhang, Q. Zhu, H. Dong, M. Zhao, A.R. Oganov, Phagraphene: a low-energy graphene allotrope composed of 5–6–7 carbon rings with distorted Dirac cones, Nano Lett. 15 (9) (2015) 6182–6186.
[60] H. Sun, S. Mukherjee, C.V. Singh, Mechanical properties of monolayer pentagraphene and phagraphene: a first-principles study, Phys. Chem. Chem. Phys. 18 (2016) 26736–26742.
[61] Z. Chang, R. Yang, Y. Wei, The linear-dependence of adhesion strength and adhesion range on temperature in soft membranes, J. Mech. Phys. Solids 132 (2019) 103697.
[62] J.S. Bunch, S.S. Verbridge, J.S. Alden, A.M. van der Zande, J.M. Parpia, H.G. Craighead, P.L. McEuen, Impermeable atomic membranes from graphene sheets, Nano Lett. 8 (8) (2008) 2458–2462.
[63] S. Deng, A.V. Sumant, V. Berry, Strain engineering in two-dimensional nanomaterials beyond graphene, Nano Today 22 (2018) 14–35.
[64] G. Gui, J. Li, J. Zhong, Band structure engineering of graphene by strain: first-principles calculations, Phys. Rev. B 78 (2008) 075435.
[65] G. Cocco, E. Cadelano, L. Colombo, Gap opening in graphene by shear strain, Phys. Rev. B 81 (2010) 241412.
[66] B. Ram, A.K. Singh, Strain-induced indirect-to-direct band-gap transition in bulk SnS_2, Phys. Rev. B 95 (2017) 075134.
[67] S.B. Desai, G. Seol, J.S. Kang, H. Fang, C. Battaglia, R. Kapadia, J.W. Ager, J. Guo, A. Javey, Strain-induced indirect to direct bandgap transition in multilayer WSe_2, Nano Lett. 14 (8) (2014) 4592–4597.
[68] B. Ram, A. Manjanath, A.K. Singh, Simultaneous tunability of the electronic and phononic gaps in SnS_2 under normal compressive strain, 2D Mater. 3 (1) (2016) 015009.
[69] W.S. Yun, S.W. Han, S.C. Hong, I.G. Kim, J.D. Lee, Thickness and strain effects on electronic structures of transition metal dichalcogenides: 2h-MX_2 semiconductors (M =Mo, W; X S=, Se, Te), Phys. Rev. B 85 (2012) 033305.
[70] D. Akinwande, C.J. Brennan, J.S. Bunch, P. Egberts, J.R. Felts, H. Gao, R. Huang, J.-S. Kim, T. Li, Y. Li, K.M. Liechti, N. Lu, H.S. Park, E.J. Reed, P. Wang, B.I. Yakobson, T. Zhang, Y.-W. Zhang, Y. Zhou, Y. Zhu, A review on mechanics and mechanical properties of 2d materials—graphene and beyond, Extreme Mech. Lett. 13 (2017) 42–77.
[71] J. Zhang, X.Y. Lang, Y.F. Zhu, Q. Jiang, Strain tuned InSe/MoS_2 bilayer van der waals heterostructures for photovoltaics or photocatalysis, Phys. Chem. Chem. Phys. 20 (2018) 17574–17582.
[72] E. Westsson, S. Picken, G. Koper, The effect of lattice strain on catalytic activity, Chem. Commun. 55 (2019) 1338–1341.
[73] B. Adolph, V.I. Gavrilenko, K. Tenelsen, F. Bechstedt, R. Del Sole, Nonlocality and many-body effects in the optical properties of semiconductors, Phys. Rev. B 53 (1996) 9797–9808.